

Date	30 June 2026	Maximum Marks	2 Marks
Team ID		Project Name	Strategic Product Placement Analysis System

Project Design Phase

Problem - Solution Fit Template

Problem - Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why.

Purpose:

- Solve complex problems in a way that fits the state of your customers.
- Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- Sharpen your communication and marketing strategy with the right triggers and messaging.
- Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- Understand the existing situation in order to improve it for your target group.

1 Customer Segment(s)

Who is your customer?

- **Retail Merchandisers:** Corporate planners deciding macro-level store layouts.
- **Retail Data Analysts:** Tasked with finding ROI in physical shelf space.
- **Store Managers:** Executing local cross-merchandising displays.

4 Emotions: Before/After

How do customers feel before and after?

- **Before:** Overwhelmed by raw data; frustrated by "guessing"; blind to physical traffic patterns.
- **After:** Confident in data-driven layouts; empowered by clear visualizations; in control.

2 Jobs-to-be-Done / Problems

Which jobs-to-be-done do you address?

- **Identify Hotspots:** Determine optimal physical locations to cross-merchandise premium products.
- **Disrupt Autopilot:** Strategically place items to break shopper routine and increase impulse buys.
- **Data Synthesis:** Translate overwhelming POS data into actionable 3D/2D spatial context.
- **Prove ROI:** Justify premium shelf-space negotiations with concrete traffic metrics.

10 Your Solution

How do you solve this within constraints?

- **Hardware-Free Spatial Engine:** Ingests standard 2D store floor plans (PDF/PNG) and POS CSV data.
- **AI Heatmapping:** Algorithms map sales density to physical aisles, generating an interactive spatial heatmap overlay.
- **Predictive Placements:** Highlights underutilized "cold zones" adjacent to "hot zones" perfect for displays.
- **Exportable Strategy:** Generates PDF layout guides for store managers to implement physical display changes.

9 Problem Root Cause

What is the real reason this problem exists?

- **Physical retail environments** lack the granular "click-tracking" analytics found in e-commerce.
- Mentally mapping thousands of rows of 2D sales data into a physical 3D space is cognitively impossible for humans at scale.

5 Available Solutions

What are the current alternatives?

- **Hardware Sensors/Cameras:** Highly accurate but require massive CapEx and cause privacy concerns.
- **Third-party Consultants:** Provide point-in-time analysis but are extremely expensive and slow.
- **Trial & Error:** Moving displays manually and tracking sales over months (slow feedback loop).

3 Triggers

What triggers customers to act?

- Introduction of a new premium product line requiring maximum physical visibility.
- Declining physical store sales compared to e-commerce.
- Mandates from corporate to increase average basket size per shopper.

6 Customer Constraints

What prevents them from taking action?

- **Budget:** No capital approved for installing smart-cameras or IoT shelf sensors.
- **Technical Skill:** Store managers lack training to use complex spatial software (e.g., AutoCAD).
- **Compliance:** Strict fire-safety and corporate brand rules limit display placements.

7 Behaviour

What do they currently do to address it?

- Manually export POS data into Excel and use Pivot Tables to guess which categories drive foot traffic.
- Physically walk the store floor with a clipboard, intuitively mapping shopper routing paths.

8 Channels of Behaviour

Where do they take action?

- **8.1 Online:** Researching retail tech on LinkedIn; searching for "heatmap software without cameras."
- **8.2 Offline:** Attending physical retail trade shows (e.g., NRF Big Show); physical store audits.

References:

- <https://www.ideahackers.network/problem-solution-fit-canvas/>
- <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>